INCH-POUND
MIL-DTL-24231/29A(SH)
26 April 2013
SUPERSEDING
MIL-C-24231/29(SH)
15 May 1987

DETAIL SPECIFICATION SHEET

CONNECTOR, PREFORM MOLDED INSERTS, FOR MOLDED PLUGS, TYPE I

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24231.

REQUIREMENTS:

1. All dimensions are in inches.
2. Tolerances shall be as follows, or as stated:

a. Decimals – ±0.005 inch.

b. Fractions – ±1/64 inch.

c. Angular dimensions – ±½ degree.

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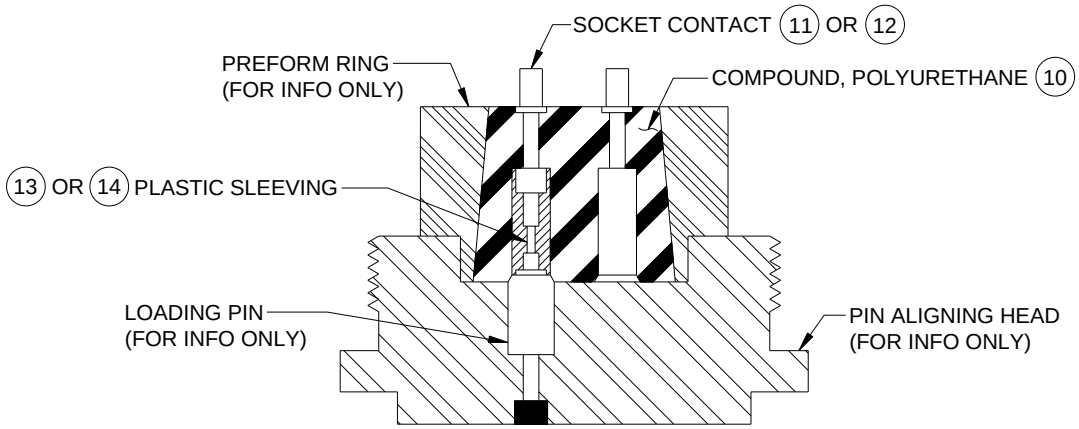


FIGURE 1. Preform insert assembly, typical.

TABLE I. Insert assembly.

Item no.	Name	Use with plug	Symbol no.
1	Insert Assembly	M24231/1-001 M24231/2-001	713 713.5
2		M24231/1-002 M24231/2-002	713.6 713.8
3		M24231/23-001	713.9
4		M24231/3-001	713.1
5		M24231/3-002	713.7
6		M24231/4-001 M24231/4-006	713.2
7		M24231/4-002 M24231/4-007	713.3
8		M24231/4-003 M24231/4-008	713.4
9		M24231/4-004 or M24231/4-005 M24231/4-009	713.10 -------- --------
15		M24231/1-003 M24231/2-003	-------- --------

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ITEM NO.	DIA A	DIA B	DIA C	DIA D	DIM E	DIM F	DIM G	DIM H
1	0.750	0.625	0.813	0.875	0.156	0.077	0.135	0.484

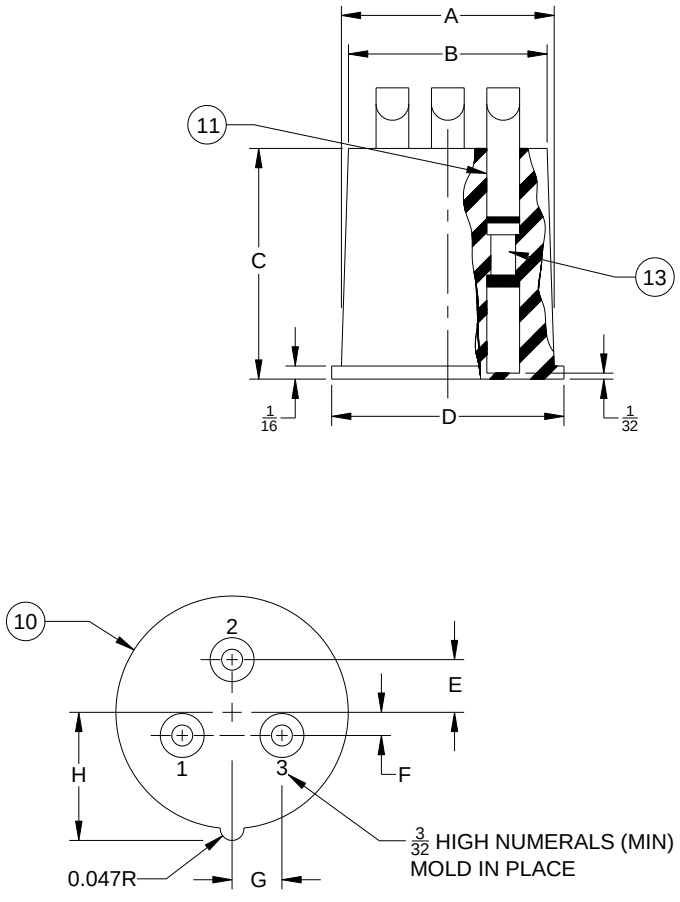


FIGURE 2. Insert assembly (3-pin). (1)

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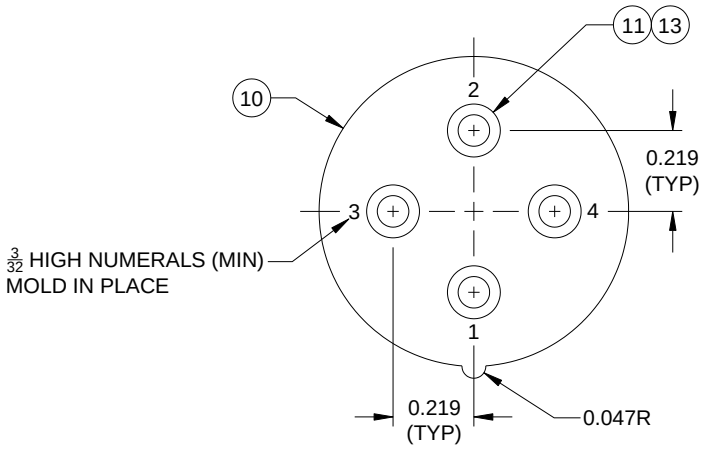


FIGURE 3. Insert assembly (4-pin). (2)
(Same as item 1, except as shown.)

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ITEM NO.	DIA A	DIA B	DIA C	DIA D	DIM E	DIM F	DIM G	DIM H
3	1.063	0.875	1.188	1.188	0.270	0.135	0.233	0.641

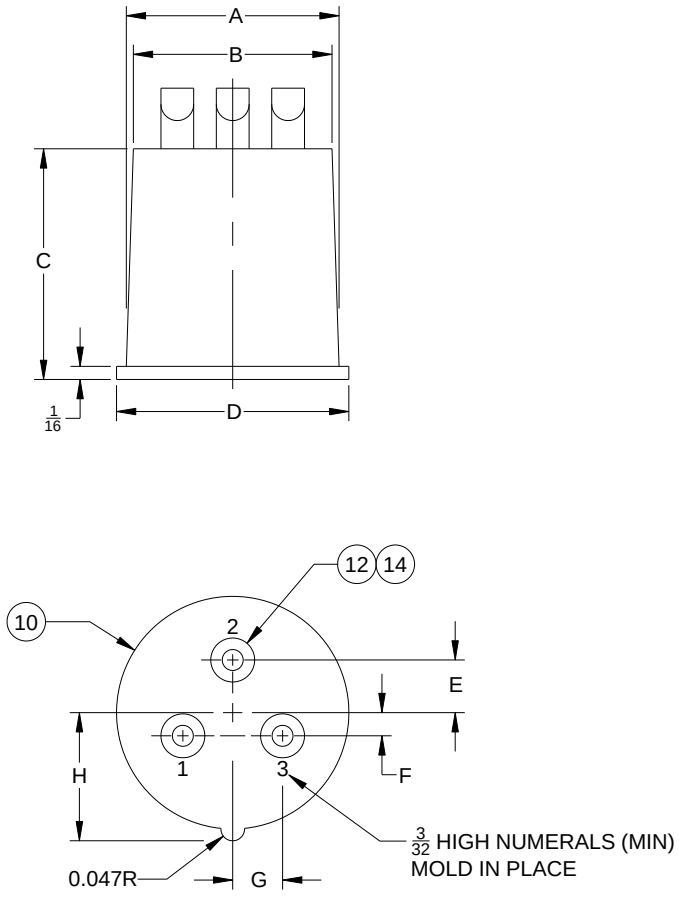


FIGURE 4. Insert assembly (3-pin, no. 12). (3)

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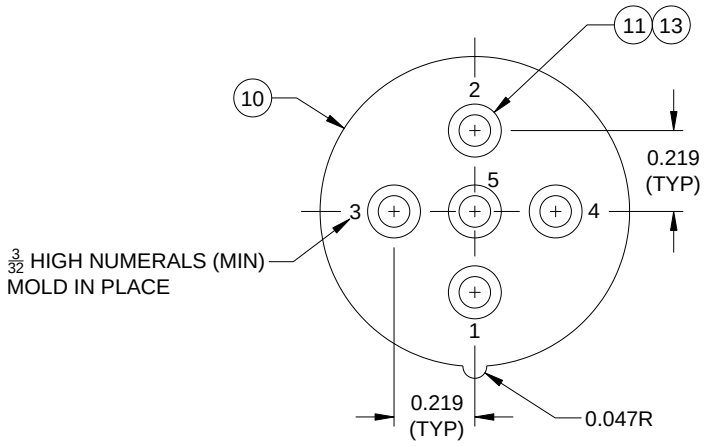


FIGURE 5. Insert assembly (5-pin). (15)
(Same as item 2, except as shown.)

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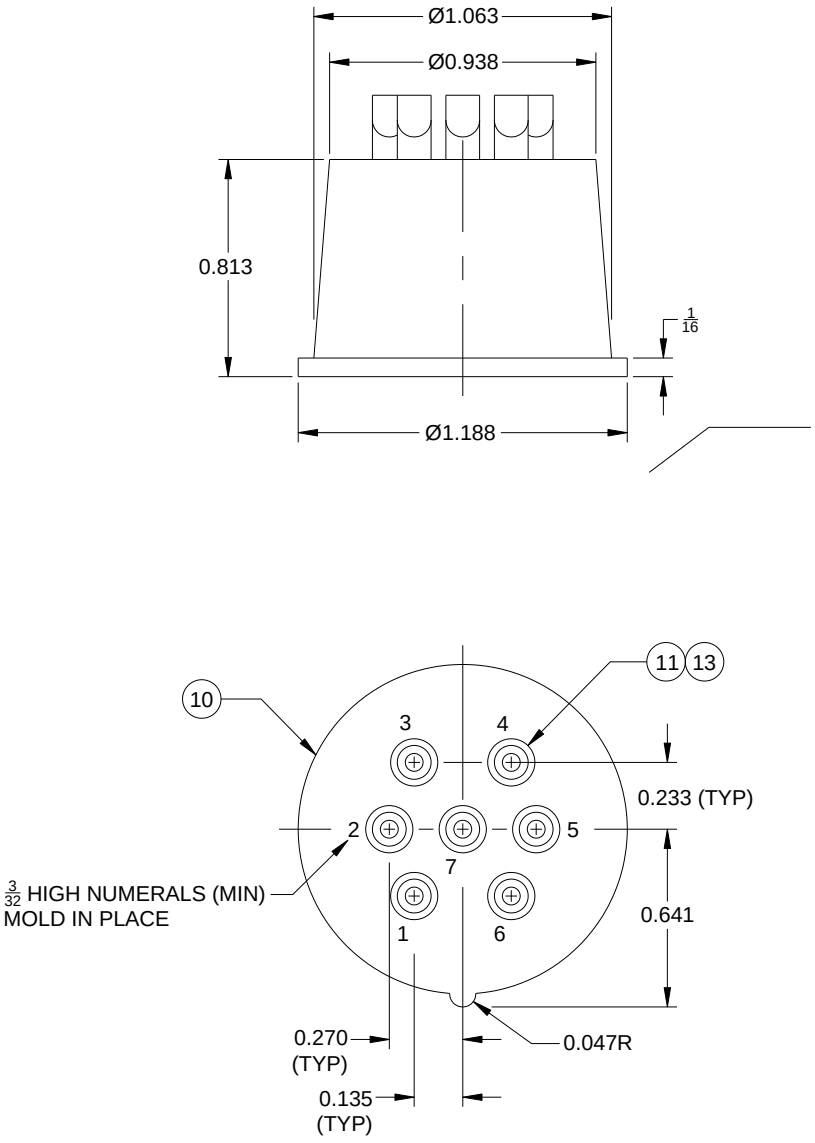


FIGURE 6. Insert assembly (7-pin). (4)

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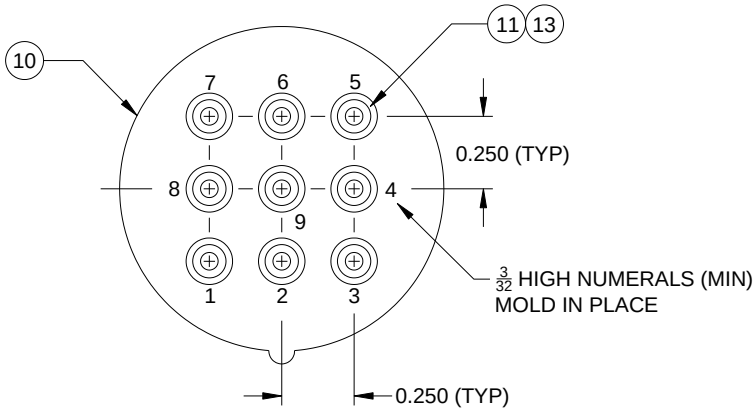


FIGURE 7. Insert assembly (9-pin). (5)
(Same as item 4, except as shown.)

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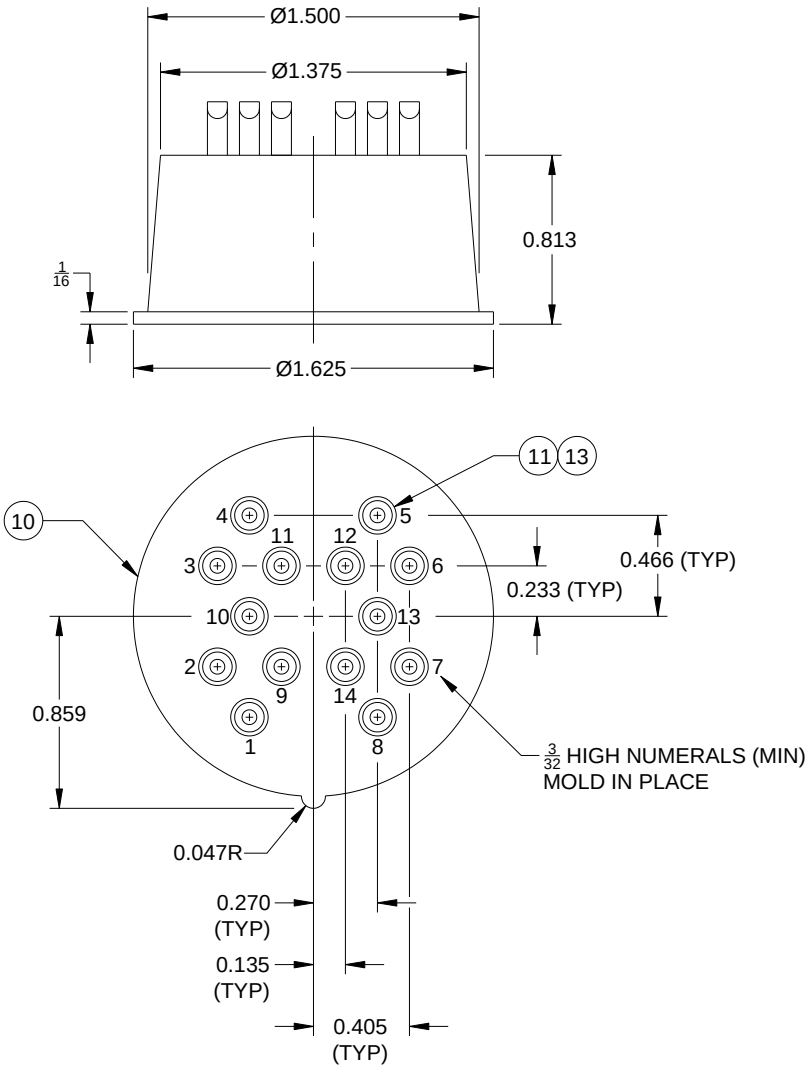


FIGURE 8. Insert assembly (14-pin). (6)

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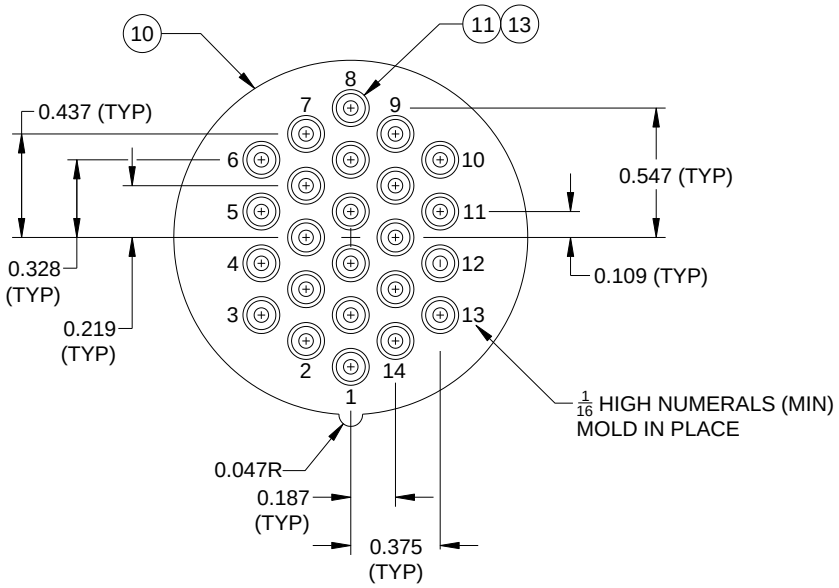


FIGURE 9. Insert assembly (24-pin). (7)
(Same as item 6, except as shown.)

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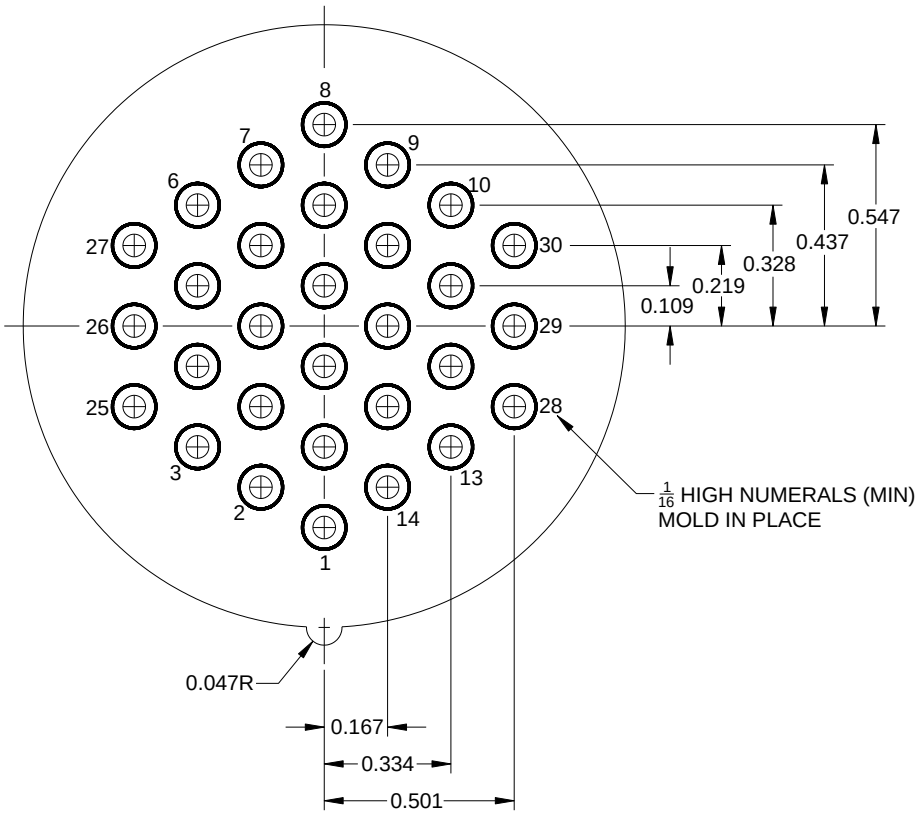


FIGURE 10. Insert assembly (30-pin). (8)
(Same as item 6, except as shown.)

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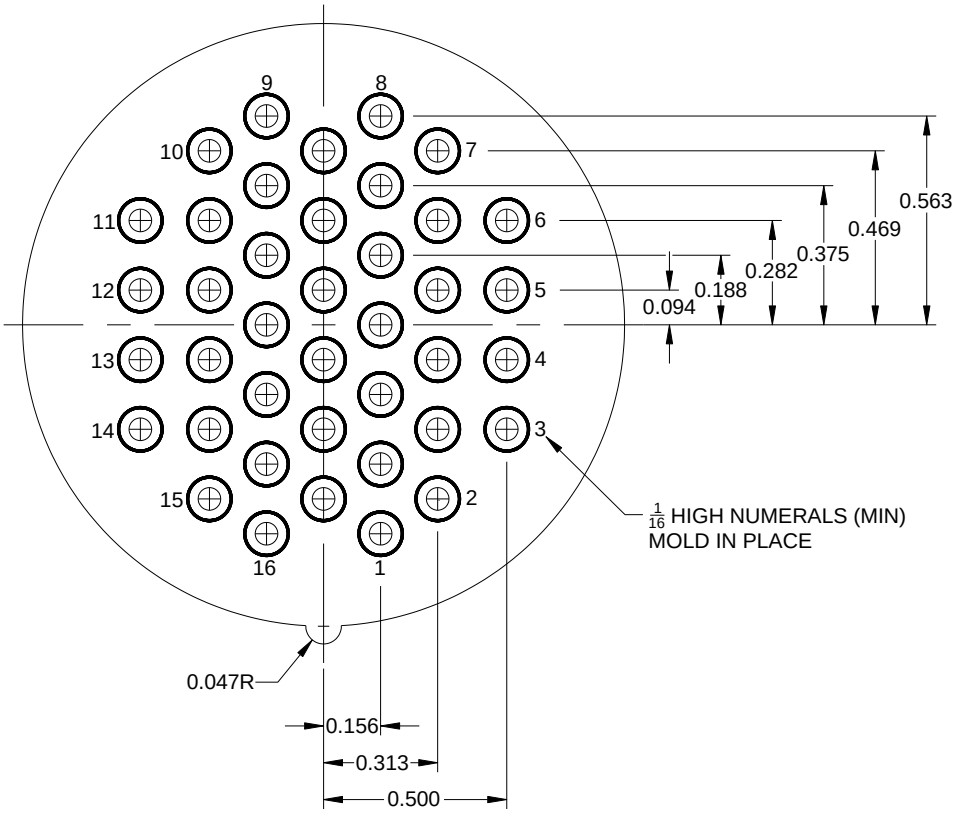


FIGURE 11. Insert assembly (40-pin). (9)
(Same as item 6, except as shown.)

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REQUIREMENTS (CONTINUED):

1. The molded inserts shall be used when molding symbol numbers 713, 713.1, 713.2, 713.3, 713.4, 713.5, 713.6, 713.7, 713.8, 713.9, and 713.10 plugs – plus the 5-conductor plugs conforming to the applicable specification sheet of MIL-DTL-24231 (see table I).

2. Design. Dimensions and configuration: See figure 1 through figure 11. Locating dimension for socket contacts for reference only, since the location will be controlled by the applicable pin aligning head.

3. Materials. As specified in table III.

4. General information. Assignment of military part numbers and general information shall be as follows and as shown in table II.

TABLE II. Assignment of military part numbers and general information.

Part no.	Use with symbol no.	Description
M24231/29-001	713 and 713.5	3-conductor straight or 90 degrees angled
M24231/29-002	713.6 and 713.8	4-conductor straight or 90 degrees angled
M24231/29-003	713.9	3-conductor (12-pin)
M24231/29-004	713.1	7-conductor
M24231/29-005	713.7	9-conductor
M24231/29-006	713.2	14-conductor straight or 90 degrees angled
M24231/29-007	713.3	24-conductor straight or 90 degrees angled
M24231/29-008	713.4	30-conductor straight or 90 degrees angled
M24231/29-009	713.10	40-conductor straight or 90 degrees angled
M24231/29-010	None	5-conductor straight or 90 degrees angled

a. Contact surfaces of socket contacts, items 11 and 12, shall be free from dirt, grease, or any other contaminants at all times before, during, and after molding.

b. Sleeving shall be installed over each socket with end of sleeve 0.031 inch from face of plug. Fit shall be snug and precautions shall be taken to prevent entrance of polyurethane into socket contact and spring clip area.

c. Solder cups shall be aligned to facilitate soldering. Surfaces of all socket contacts shall be free of any molding compound or any other contaminant. Removal of contaminants after molding shall not be allowed as cleaning may damage or remove gold plating.

d. Insert assemblies, items 1 through 9 and 28, shall be molded as follows:

(1) Slip plastic sleeving over socket contacts.

(2) Assemble socket contacts on loading pins in pin aligning head.

(3) Push the plastic sleeving down until sleeve rests on shoulder of loading pin.

(4) Place preform ring on pin aligning head, as shown on figure 1, and fill the area around the socket contacts with molding compound, item 10, to top of preform ring.

(5) Allow to stand until all entrapped air rises to the surface, approximately 15 minutes. Refill to top of preform ring, if necessary. Keep solder cups clean.

(6) Place assembly into 180 °F oven and cure for 3 hours.

(7) After curing, allow assemblies to cool until preform molded insert can be removed from preform ring and pin aligning head.

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TABLE III. Parts list.

Item no. 1/	Description 2/	Quantity for M24231/29- 3/										Material/specification	Remarks
		001	002	003	004	005	006	007	008	009	010		
1	Insert assembly	1										See table I.	See figure 1 and figure 2.
2	Insert assembly		1									See table I.	See figure 1 and figure 3.
3	Insert assembly			1								See table I.	See figure 1 and figure 4.
4	Insert assembly				1							See table I.	See figure 1 and figure 6.
5	Insert assembly					1						See table I.	See figure 1 and figure 7.
6	Insert assembly						1					See table I.	See figure 1 and figure 8.
7	Insert assembly							1				See table I.	See figure 1 and figure 9.
8	Insert assembly								1			See table I.	See figure 1 and figure 10.
9	Insert assembly									1		See table I.	See figure 1 and figure 11.
10	Compound, molding	As required										4/	See figure 1.
11	Contact, socket, size 16	3	4		7	9	14	24	30	40	5	5/	See figure 1 and figure 2.
12	Contact, socket, size 12			3								5/	See figure 1 and figure 4.
13	Sleeving, size 9	3	4		7	9	14	24	30	40	5	Plastic, MIL-I-631 6/	See figure 1 and figure 2.
14	Sleeving, size 6			3								Plastic, MIL-I-631 6/	See figure 1 and figure 4.
15	Insert assembly										1	See table I.	See figure 1 and figure 5.

NOTES:

1/ Item numbers are the same as encircled numbers shown on figures.

2/ See applicable figure or table listed in remarks column for part numbers and dimensions.

3/ Quantity is for one complete assembly of each size.

4/ Molding compound to be procured in accordance with S9320-AM-PRO-020.

5/ Socket contact, item 11, shall be Amphenol Corp. size 16M, short, part no. 10-101515-15S, gold plated; L.L. Rowe Company, part no. 40097; or equal. Socket contact, item 12, shall be Amphenol, size 12, solder type, part no. 10-33644, silver plated, followed by gold plate, not less than 0.0001 inch thick; or equal. Any other socket contact submitted as equal shall be reviewed by NAVSEA.

6/ Plastic sleeving, items 13 and 14, shall be in accordance with Type F, Form U_a, Grade a, Class II, Category I, color transparent (clear) of MIL-I-631.

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 5935-2011-078)

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